

# HS Data Science 26

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## Session 02: Scientific Working I — The Research Loop

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# Session Overview

## Module A — The Research Loop

Steps 1–11 of the scientific cycle:

- from reading literature to communicating results
- hypothesis as the contract with the reader
- three experiment modes: dev loop · pre-experiment · full

## Module B — Experiment Design

- conditions, controls, and ablations
- choosing and applying the right metric
- pitfalls that invalidate results

## Why this session?

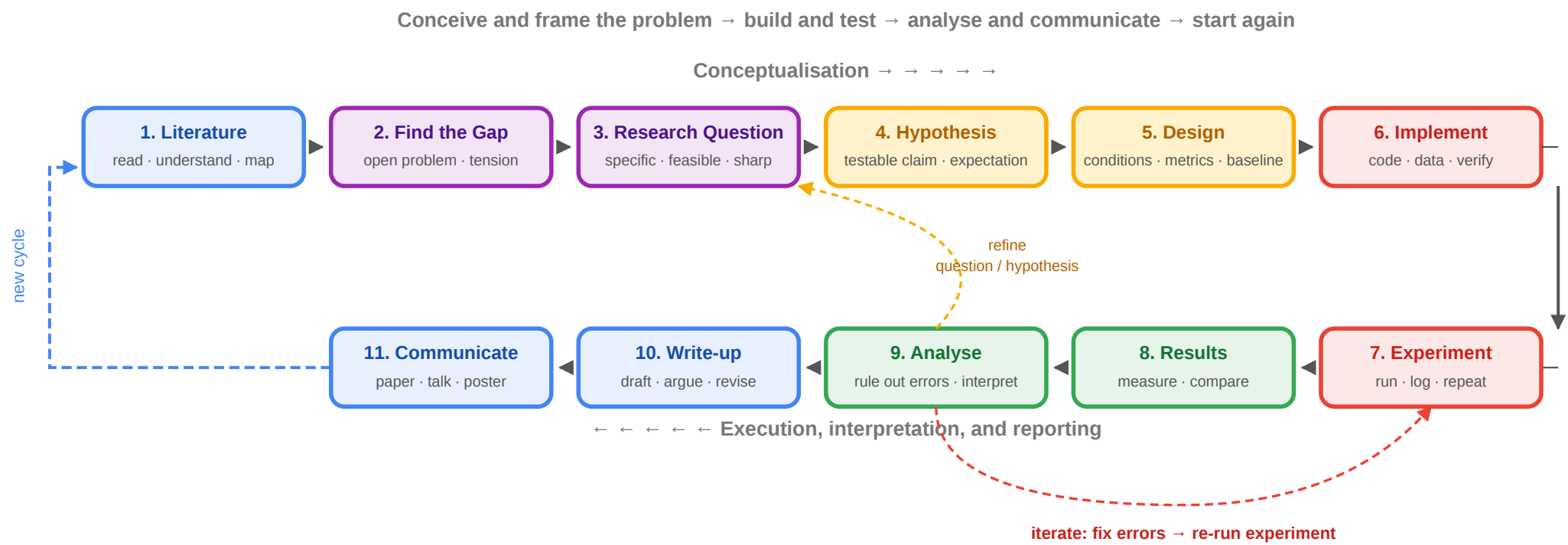
Your seminar project is a *mini research loop*. Every step applies — even at small scale.

**The key insight:** A well-designed experiment with a modest model beats a sophisticated model with a broken experiment every time.

# The Research Loop

**Guiding Question:** What steps turn a vague question into a publishable result?

# The Research Loop — Overview



Science is iterative: framing, testing, analysis, and communication all feed the next cycle.

# Step 1 — Reading Literature

The three-pass method (Keshav 2007)

| Pass | What you read               | Time      |
|------|-----------------------------|-----------|
| 1st  | title, abstract, conclusion | 5 min     |
| 2nd  | intro, headings, figures    | 20 min    |
| 3rd  | full paper, related work    | 60–90 min |

Only go to pass 3 for **directly relevant** papers.

**While reading, ask:**

- What claim does this paper make?
- What evidence supports it?
- What does it assume or leave out?
- What would a follow-up study do?

**Active reading — example questions**

- *"Could this result be a confound?"*
- *"What changes if the dataset is different?"*
- *"Does this generalise beyond the benchmark?"*

**Keep a reading log:** one sentence per paper — the claim and why you care. Use Obsidian or Notion. Build this habit now.

Reading is not collecting — it is  
**understanding what is settled and what  
remains open.**

# Step 2 — Finding the Gap

Types of gap *(be complete — gaps fall into these categories)*

| Type                | Example signal                      |
|---------------------|-------------------------------------|
| Missing condition   | "Only tested on English"            |
| Conflicting results | "Papers A and B disagree"           |
| Missing baseline    | "No comparison to simple heuristic" |
| Scope limitation    | "Offline only, not online"          |
| Scalability limit   | "Fails above 1M documents"          |
| Methodological flaw | "No significance test"              |

Where to look:

- "Limitations" and "future work" sections
- hedging language: *"we suspect", "future work should"*
- points of disagreement across papers

A gap is not "nobody did this".

A gap is a *tension, a limit, or an open question that matters* — and that is investigable within your constraints.

From gap to research question:

*"Does query reformulation improve answer quality in agentic search on the Open Web Index?"*

The gap drives the question. The question drives the hypothesis. The hypothesis drives everything else.

Ask: *"What would a practitioner need that the literature does not yet provide?"*

# Step 3 — The Research Question

## From vague to sharp

| Too vague              | Sharp                                                                              |
|------------------------|------------------------------------------------------------------------------------|
| "Is RAG better?"       | "Does BM25 re-ranking improve answer faithfulness on TriviaQA vs. dense-only RAG?" |
| "Does my system work?" | "Does my system outperform a keyword-search baseline on OWI Curlie using nDCG@10?" |

A sharp question names: **method · dataset · metric · comparison.**

## Criteria for a good research question

- **Specific** — names method, dataset, and metric
- **Empirical** — requires evidence, not just reasoning
- **Interesting** — the answer changes how someone acts
- **Testable** — we can design an experiment to answer it
- **Feasible** — we have the resources and time to run it

**Testable ≠ Feasible.** *Testable*: can we construct the right experiment? *Feasible*: do we have the compute, data, and time? Both must be true.

# Step 4 — The Hypothesis

## Hypothesis template:

*"If [intervention], then [outcome], measured by [metric], on [data], compared to [baseline]."*

## Example:

*"If we apply query reformulation before dense retrieval, nDCG@10 improves by  $\geq 5\%$  vs. direct dense retrieval on OWI Curlie queries."*

## Falsifiability:

*You must be able to run an experiment that proves you wrong.*

If no result could disprove the claim, it is a belief, not a hypothesis.

## Properties to verify

| Property          | Check                                 |
|-------------------|---------------------------------------|
| Falsifiable       | What result would disprove it?        |
| Directional       | Does it predict a specific direction? |
| Operationalisable | Can you measure it?                   |
| Feasible          | Testable within your timeframe?       |
| Grounded          | Does related work make it plausible?  |

## Anti-patterns

- "The system will be better" ← not directional
- "LLMs are intelligent" ← not falsifiable
- "Performance improves" ← no metric, no baseline



# Step 5 — Experiment Design

## Design before you code

| Element    | Decision to make                       |
|------------|----------------------------------------|
| Conditions | which system variants to compare       |
| Baseline   | simplest reasonable alternative        |
| Dataset    | which data, which split, held-out test |
| Metric     | primary + one secondary                |
| Protocol   | how many runs, how to average          |
| Resources  | GPU-hours, storage, API cost budget    |

**Check resources before committing.** A full experiment that exceeds your compute budget is a wasted design.

## Pre-registration

Write down hypothesis + design *before* seeing results. This prevents:

- p-hacking
- HARKing (Hypothesising After Results are Known)
- post-hoc cherry-picking

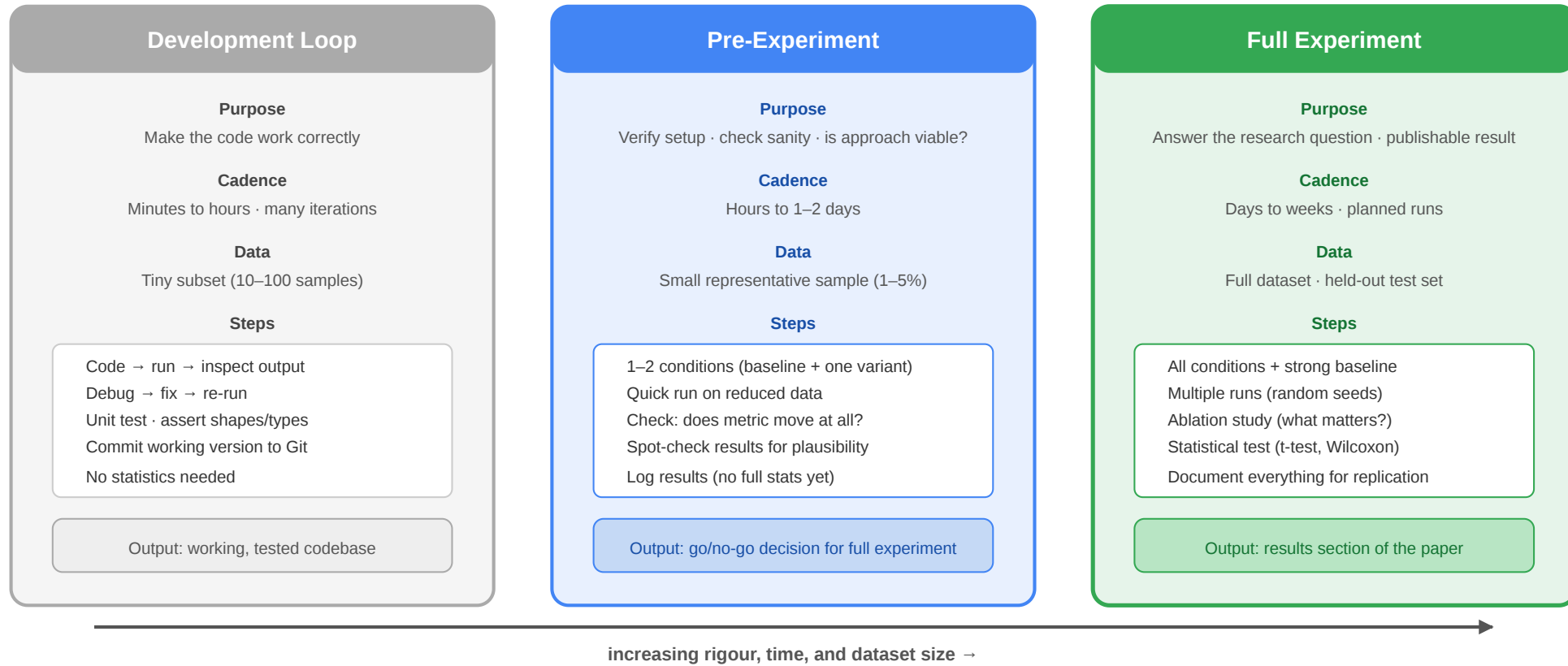
## Recommendation

1. One strong baseline
2. One primary condition
3. One ablation
4. Fixed train / dev / test split — no leakage
5.  $\geq 3$  runs with different seeds

A one-paragraph design document written before running is worth more than any metric improvement.

# Steps 6 & 7 — Three Experiment Modes

## Three Experiment Modes in AI/ML Research



# Step 6 — Implementation

## The development loop mindset

Work in small, verifiable steps:

1. code → run → inspect output
2. assert shapes, types, and ranges
3. overfit one batch deliberately — if you can't, the pipeline is broken
4. commit working state to Git after each milestone

## Version control is not optional

```
feat: BM25 baseline on MSMARCO dev
```

```
MRR@10 = 0.187 - matches Thakur et al. 2021.
```

```
Pipeline confirmed correct. Next: dense retrieval.
```

Each commit message is a lab notebook entry.

## Environment and reproducibility

- pin dependencies (`requirements.txt` or `pyproject.toml`)
- fix random seeds at the top of every script
- keep data preprocessing in code — never edit raw data manually
- log every hyperparameter to a config file, not inline

## Sanity checks before scaling up:

- Does random perform worse than your model?
- Does overfitting one batch produce near-zero loss?
- Do evaluation scores on dev match your expectation from related work?

# Step 7 — Running Experiments · Step 8 — Results

## Running experiments systematically

- one experiment = one folder with `config.yaml` + `results.json` + `README.md`
- log **all** runs — including the ones you will not report
- run every condition with  $\geq 3$  seeds
- measure wall-clock time and GPU memory — part of your resource budget

## From raw numbers to reported results

1. Compute mean  $\pm$  std across seeds
2. Compare conditions on the **held-out test set** — **once**
3. Do not re-run on test after seeing results
4. Plot distributions, not just means

## Honest reporting checklist

- ☐ All conditions reported, not just the best
- ☐ Held-out test used only once
- ☐ Variance reported (not just mean)
- ☐ Baseline is as strong as you could make it
- ☐ No data leakage in the split
- ☐ Statistical significance tested

# Step 9 — Ruling Out Errors

## Common error types

| Error              | What it looks like                       |
|--------------------|------------------------------------------|
| Data leakage       | test items in training context           |
| Wrong metric       | metric $\neq$ what the hypothesis claims |
| Evaluation bug     | off-by-one, wrong split, wrong average   |
| Confound           | model memorises dataset artefacts        |
| Weak baseline      | unfair comparison                        |
| Random variance    | one lucky seed drives the result         |
| Distribution shift | train/test from different sources        |

## Debugging process

1. Sanity-check: does random perform worse?
2. Overfit one batch on purpose
3. Inspect 20–30 model outputs manually
4. Ablate one component at a time
5. Re-run with 3 different seeds
6. Trace train/dev/test construction end-to-end

**If results are surprisingly good:** Stop and verify. Unexpectedly good results are more suspicious than bad ones.

Rule out errors *before* writing up — not during the reviewer response.



# Step 9b — Limitations

## What a limitation is

A limitation is a boundary on what your results can claim — not an excuse, not a weakness to hide. Every study has them. Naming them honestly is a mark of scientific maturity and protects you from overreach.

## Types of limitations

| Type                 | Example                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------|
| Dataset scope        | Evaluated on English web text only — findings may not transfer to other languages or domains |
| Evaluation protocol  | Automatic metrics (BLEU, NDCG) may not align with human judgement                            |
| Threat to validity   | Participants were self-selected; results may not generalise to the full population           |
| Computational budget | Only one model size tested; scaling behaviour unknown                                        |



## How to write a limitations section

1. **Be specific** — "our dataset may not be representative" is not enough. State *which* aspect is limited and *why* it matters for your conclusions.
2. **Connect to claims** — link each limitation to the research question it affects. Does it change the direction of your finding, or only its magnitude?
3. **Do not apologise** — limitations frame the scope of validity; they are not confessions.
4. **Distinguish limitations from future work** — a limitation constrains the current result; future work is what you would do to lift the constraint.

**In your seminar narrative:** identify at least one concrete limitation of your planned approach *before* you run any experiments.

| Type               | Example                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------|
| Construct validity | "Relevance" is operationalised as click-through — this captures one dimension of relevance, not all |

Early acknowledgement prevents overreach in the final report.



# Steps 10–11 — Write-up and Communicate

The paper is an argument, not a report

| Section      | Role                                   |
|--------------|----------------------------------------|
| Abstract     | claim + evidence + impact in 150 words |
| Introduction | motivation + gap + your contribution   |
| Related Work | what exists and why it falls short     |
| Method       | what you did and why                   |
| Experiments  | how you tested it                      |
| Results      | what the evidence shows                |
| Analysis     | what it means and its limits           |
| Conclusion   | restate claim + future work            |

**Write the abstract last** — once you know what you found.

## Consolidation checklist

- ☐ One central claim — everything supports it
- ☐ Every figure caption states the takeaway
- ☐ Limitations section is honest
- ☐ Code and config available for reproduction

## Communication formats

- **Talk:** 12-minute argument — one key idea
- **Poster:** one question, one answer, one figure
- **Report:** ACM format, max 8 pages

Say one thing clearly. Scientists remember papers that make a single sharp point.

# Experiment Design in AI/ML

**Guiding Question:** How do you design an experiment that actually answers your research question?

# Experiment Design — Overview

## Topics in this module

- conditions, controls, and ablation studies
- evaluation metrics — choosing and applying them correctly
- common pitfalls that invalidate results
- statistical significance and effect size

**Practical goal:** After this module you can design a minimal but rigorous experiment for your seminar project — on paper, before writing a line of code.

# Conditions, Controls, and Ablations

## Definitions

| Term      | Meaning                                |
|-----------|----------------------------------------|
| Control   | unchanged reference — the baseline     |
| Condition | a variant differing in one factor      |
| Ablation  | full system <i>minus</i> one component |

## Example — RAG pipeline

- Baseline: BM25 + GPT-4
- Condition A: Dense retrieval + GPT-4
- Condition B: Dense + BM25 hybrid + GPT-4
- Ablation: B without re-ranking
- Ablation: B without query expansion

Each ablation isolates one contributing factor.

## Design rule: one variable at a time

If Condition B differs from A in *both* retrieval method *and* prompt format, you cannot isolate the cause of any difference.

Change one thing at a time. When unavoidable, document the confound explicitly.

## Minimum viable experiment

1. One baseline (simple, strong, cited)
2. One primary condition (your system)
3. One ablation (system minus your key component)
4. Two metrics (primary + secondary)
5. Fixed dataset split



Ablations are often more convincing than the main result — they show *what* drives the improvement.

# Metrics and Evaluation

## The core principle

The metric must directly measure the property your hypothesis claims. Derive the metric from the hypothesis — never the reverse.

**Common metrics by task** (*representative, not exhaustive*)

| Task               | Metric                        |
|--------------------|-------------------------------|
| Retrieval (ranked) | nDCG@k, MAP, Recall@k         |
| Classification     | F1 (macro/micro), MCC         |
| Generation         | ROUGE, BERTScore, BLEURT      |
| Faithfulness (RAG) | FactScore, Ragas faithfulness |
| Human eval         | Likert, pairwise preference   |

## Metric traps

| Trap                        | Problem                          |
|-----------------------------|----------------------------------|
| BLEU for faithfulness       | measures overlap, not factuality |
| Accuracy on imbalanced data | majority class trivially wins    |
| Mean without variance       | one lucky run inflates results   |
| Proxy without validation    | high proxy, low human preference |

## Dataset quality matters as much as the metric

Before using a benchmark, check:

- How was it constructed? By whom?
- Does it have known biases or artefacts?
- Are the labels reliable (inter-annotator agreement)?

A biased dataset with a correct metric still

# Common Pitfalls in ML Research

## Data pitfalls

- **Train/test overlap** — shuffling before split leaks test items into training
- **Temporal leakage** — future data used to predict past events
- **Benchmark overfitting** — tuning on the test set, even unintentionally
- **Label noise** — gold labels wrong in 5–15% of NLP datasets
- **Dataset bias** — benchmark constructed from a specific source generalises poorly

## Resource pitfalls

- **Insufficient compute** — full experiment requires GPU-hours not available
- **Storage underestimated** — embeddings, checkpoints, logs exceed quota



## Model and reporting pitfalls

- **Spurious correlations** — model learns dataset artefacts, not the task
- **LLM-as-judge trained on same distribution** — evaluator is biased toward the model it judges
- **p-hacking** — try many metrics, report only the significant one
- **HARKing** — present post-hoc hypothesis as pre-specified
- **Selective ablations** — run 10, report 3
- **No error bars** — single run, no variance

## Structural safeguards

1. Write hypothesis *before* seeing results
2. Commit evaluation script to Git *before* running on test set
3. Log every run — runs you don't report still happened

- **API cost overrun** — LLM calls at scale cost 10× the estimate

Estimate resources before running. A design that cannot be executed is not a design.

4. Report your worst result alongside your best

Pre-registration turns pitfalls into choices — you have to actively decide to violate them.

*Source: Pineau et al. 2021; Dodge et al. 2019*



# Statistical Significance and Effect Size

## Statistical tests for ML

| Situation                            | Test                          |
|--------------------------------------|-------------------------------|
| Two systems, paired samples          | Wilcoxon signed-rank          |
| Two classifiers, same test set       | McNemar's test                |
| Means over multiple runs             | Paired t-test ( $n \geq 30$ ) |
| Multiple conditions vs. one baseline | Bonferroni correction         |

**Threshold:**  $p < 0.05$  conventional, but:

- report the exact p-value
- $p = 0.049$  and  $p = 0.051$  are not meaningfully different
- apply multiple-test correction when comparing many conditions

## Effect size matters more than p

| Measure                    | Use                          |
|----------------------------|------------------------------|
| Cohen's d                  | standardised mean difference |
| $\Delta$ metric (absolute) | nDCG +0.03 vs. +0.003        |
| Relative improvement       | "+5% over baseline"          |

## Practical significance check

Even a significant +0.001 nDCG is rarely worth deploying. Ask:

- Is it reproducible on other datasets?
- Is it consistent across subgroups?
- Would a user notice the difference?

Significant but tiny may matter less than non-significant but large and consistent.

# Wrap-Up



## The Research Loop

- 11 steps: Literature → Gap → Research Q. → Hypothesis → Design → Implement → Experiment → Results → Analyse → Write-up → Communicate → repeat
- Iteration is not failure — it is the mechanism of science
- Steps 1–5 determine quality more than steps 6–8

## Experiment Design

- Conditions, controls, ablations: one variable at a time
- Metric follows from hypothesis — never the reverse
- Pre-register: design before you see results
- Estimate resources before you commit to a design

## For your project — do this now:

1. Write your research question in one sentence (method · dataset · metric)
2. State a falsifiable, directional hypothesis
3. Name your baseline and justify it
4. Define your primary metric and why it matches your hypothesis
5. Estimate your resource budget (compute · data · time)

**Next session (S03):** literature search in depth, reference management, the publication process, project management with GTD and Git.

Bring a written research question and hypothesis to S03. These will be reviewed as part of your research narrative (due 12/05).

## Image Credits & References

Dodge, Jesse and Gururangan, Suchin and Card, Dallas and Schwartz, Roy and Smith, Noah A. (2019). Show Your Work: Improved Reporting of Experimental Results. *Proceedings of EMNLP-IJCNLP 2019*.

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